

Fahad Bin Mohammad Hossain

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience building and operating scalable multi-tenant systems using NestJS, MongoDB, RabbitMQ, Redis etc focused on event-driven architecture, secure authentication, rate limiting, and AI/LLM integrations (RAG, Embeddings, Vector Search, Frontier Lab APIs Integrations) for production backend infrastructure at a global SaaS company.

EXPERIENCE

Associate Software Engineer II, 6sense HQ Limited - Dhaka, Bangladesh Jan 2026 - Present

- Reduced business logic execution errors by 13% by building scalable backend APIs with retry mechanisms and async workflows across interconnected modules, using NestJS, PostgreSQL, MongoDB, and modular architecture.
- Increased data pipeline throughput significantly through event-driven processing pipelines that combine Voyage AI embeddings, Claude APIs, background workers, and queue-based messaging for higher fault tolerance.
- Strengthened multi-tenant security by implementing JWT/OAuth authentication with RBAC and distinct rate limits per tenant and per role, improving authorization performance under concurrent load.
- Cut deployment time by 20% by managing containerized CI/CD workflows with multi-stage Docker and AWS that support automated rollbacks across staging and production environments.

Junior Software Engineer, 6sense Technologies - Dhaka, Bangladesh Sep 2024 - Dec 2025

- Delivered 10% faster feature rollout by developing modular RESTful services in NestJS and PostgreSQL while enforcing clean coding standards and high unit test coverage across shared modules.
- Reduced query latency by 17% on high-traffic transactional workflows by optimizing PostgreSQL and MongoDB with targeted indexing, query refactoring, and efficient aggregation pipelines.
- Cut large file upload times by 60% by introducing S3 pre-signed URLs that let the frontend upload files in chunks outside the request-response cycle, improving reliability for large payloads.
- Improved deployment consistency by 20% by creating reusable validation, authentication, and testing components paired with modular service design and containerized CI/CD pipelines.

SKILLS

Languages	C++, JavaScript, TypeScript, Python
Databases	PostgreSQL, MongoDB, Redis, VectorDB
Frameworks & Tools	NodeJS, NextJS, NestJS, FastAPI, GraphQL, Docker, RabbitMQ, AWS (EC2, ECS, S3 etc)
AI Engineering	Retrieval Augmented Generation (RAG), LLM API Integration, Prompt Engineering
Architecture	Monolith, Microservices, Event Driven Design

PROFESSIONAL PROJECTS

- Pattern50 - Your startup studio for the AI era** - NestJS, NextJS, TypeScript, MongoDB, Claude, RabbitMQ, Redis, AWS (S3/ECR/EKS), K8s, 1Password, Google Drive Integration, Zoho Books, Stripe, Codacy, JWT/OAuth, Jest, Docker
 - Developed multi-tenant JWT/RBAC auth with access levels (Global / Studio / Company), invite onboarding, refresh tokens, and role-permission scoping for studio and company portfolio management across the platform.
 - Connected Google Drive OAuth and GitHub App integrations for document and code ingestion into AI knowledge bases, with Bull-based token refresh, async embedding jobs, and credentials stored in 1Password vaults.
 - Designed a RAG pipeline with Voyage embeddings, document chunking, and MongoDB Atlas vector search, plus an OpenAI-powered report generator to turn company knowledge into scoped AI answers and reports.
 - Built a Claude AI agent with multi-round tool-calling (GitHub + Codacy) and SSE streaming chat so studios can query repositories, pull requests, workflows, and code-quality signals in real time.
 - Built Stripe billing (PaymentIntents, cards, ACH, webhooks, Socket.IO events) and Zoho Books invoicing over event driven NestJS microservices (RabbitMQ, Redis, Bull), deployed via Docker to AWS ECR/EKS with GitOps CI/CD.
- Jeter AI - AI powered dating, compatibility & matchmaking platform** - NestJS, NextJS, TypeScript, PostgreSQL, Prisma ORM, OpenAI, Redis (Valkey), AWS (S3/SES/ECS/ECR), Agora, Stripe, JWT/OAuth, Jest, Docker
 - Architected a scalable, modular NestJS backend with 20+ domain modules using clean layered architecture, dependency injection, and reusable service patterns, enabling maintainable code organization and faster feature delivery across teams.
 - Developed an AI-driven compatibility engine using OpenAI APIs and token-based embeddings to generate personalized heart, mind, and body scores along with dynamic Jeter summaries from structured user assessment responses.
 - Implemented secure multi-provider authentication supporting JWT, Google OAuth, and OTP login with authorization guards, CSRF protection, rate limiting, and security headers to ensure safe and compliant user access workflows.
 - Integrated presigned AWS S3 uploads, Agora messaging/video, Stripe payments, and Redis caching, while maintaining reliability through Jest unit testing, Swagger API documentation, and CI/CD pipelines with Jenkins and GitHub Actions.

EDUCATION

Bangladesh University of Business and Technology 2019 – 2023
Bachelor of Computer Science and Engineering — CGPA: 3.63
Relevant Coursework: OS, DSA, AI, ML, Computer Networks, Database, Discrete Mathematics, Microprocessor & Microcontroller etc.